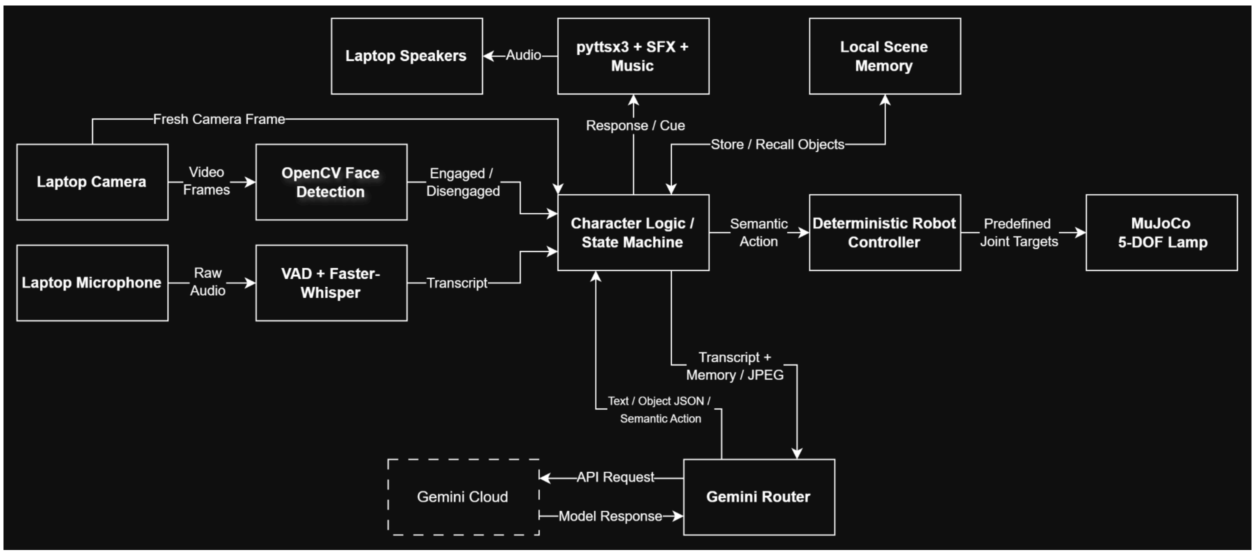
Live Character Lamp Robot - Short Technical Note

System Overview

The system provides one continuous character interaction using the laptop camera, microphone, and speaker, along with a MuJoCo-simulated lamp body. Local components handle engagement detection, voice activity, speech recognition, interaction state, audio, scene-memory storage, and deterministic motion. Gemini is used selectively for semantic scene understanding, open-ended language responses, and multimodal image-and-goal planning.

The goal was to keep latency-sensitive perception and control local while using cloud multimodal reasoning only where it provides clear value.

Architecture and Data Flow



The **Character Logic / State Machine** is the central coordinator. Camera frames feed OpenCV face detection for engagement and can also be captured for visual reasoning. Microphone audio is captured locally and transcribed by Faster-Whisper. Scene observations are stored in local structured memory.

When semantic reasoning is needed, the character logic sends the relevant transcript, memory, or compressed JPEG image through the **Gemini Router** to Gemini Cloud. Gemini returns natural-language text, structured object information, or a constrained semantic robot action.

Raw microphone audio remains local in the active pipeline. The cloud receives only text/memory or compressed camera images when required.

Model-to-Action Protocol and Simulation

- **Constrained control:** Gemini selects a semantic action (e.g., `LOOK_LEFT`, `LOOK_UP`) rather than joint angles
- **Deterministic execution:** the robot controller maps that action to predefined targets for the 5 joints, keeping motion predictable and inspectable
- **MuJoCo simulation:** the supplied URDF is animated using smooth kinematic interpolation; the emitter visually represents light state
- **Post-action check:** after a goal-directed movement, the system captures a fresh camera frame before confirming completion

Key Design Choices and Tradeoffs

- **Hybrid local/cloud**: Faster-Whisper, state management, memory, audio, and robot control run locally for responsiveness and CPU-only operation. Gemini handles higher-level visual and language reasoning, at the cost of network latency and API availability
- **Lightweight engagement**: OpenCV frontal-face detection provides a simple attention proxy. It is efficient, but less precise than true gaze estimation
- **Constrained actions**: Gemini selects from predefined semantic actions rather than joint angles. This reduces flexibility but keeps robot execution deterministic and inspectable
- **Model fallback**: The Gemini Router retries compatible models and applies cooldowns after timeouts or rate limits, improving resilience without eliminating cloud-side failures

Measurements

Engagement detection	10 / 10 trials
Disengagement detection	10 / 10 trials
Faster-Whisper transcription	~0.4 - 0.5 s
Healthy Gemini text response	~0.7s observed
Healthy multimodal planning	typically ~1 - 3 s
Cloud timeout	~10s per attempt
Python CPU usage	~3%
Python memory usage	~1.03 GB

Measured on Windows 11 with an Intel Core Ultra 7 356H and 16 GB RAM. CPU and memory values refer to the running Python process.

Deployment

The target is **Ubuntu 24.04, CPU-only**. Dependencies are listed in *requirements.txt*; the Gemini key is supplied through a local *.env* file and is not included in the submission. Faster-Whisper uses *base.en* with CPU int8 inference. Full setup instructions are in README.md.

Known Limitations and Scope

- Face detection approximates attention rather than true gaze
- Scene memory resets when the program exits
- Gemini can experience latency, rate limits, and service errors
- The laptop camera is fixed, so post-action observation is not equivalent to a moving head-mounted camera
- Robot motion is kinematic rather than actuator/torque controlled
- The simulated light changes emitter appearance but does not physically illuminate the scene

The completed system covers engagement/disengagement, speech, scene memory and recall, multimodal goal-directed action, motion, light, voice, SFX, music, post-action observation, and cloud fallback. The scope intentionally favors a coherent end-to-end interaction over broader unfinished features.